HW2

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Q1: A.29

Let $\mathbb{S} = C[0,1]$ and $d_p(f,g) \equiv \left(\int_0^1 |f(t) - g(t)|^p \, dt\right)^{\frac{1}{p}}$ for $1 \leq p < \infty$ and $d_\infty(f,g) = \sup\{|f(t) - g(t)| : t \in [0,1]\}$.

Note: May use the inequality:

$$|a+b|^p \leq 2^p \left(|a|^p + |b|^p\right)$$

(a)

Let $f(x) \equiv 1$. Let $f_n(t) \equiv 1$ for $0 \leq t \leq 1 - \frac{1}{n}$, and $f_n(t) = n(1-t)$ for $1 - \frac{1}{n} \leq t \leq 1$. Show that $d_p(f_n, f) \to 0$ for $1 \leq p < \infty$ but $d_\infty(f_n, f) \not\to 0$.

Answer

(b)

Fix $f \in C[0,1]$. Let $g_n(t) = f(t)$, $0 \leq t \leq 1 - \frac{1}{n}$, and $g_n(t) = f\left(1 - \frac{1}{n}\right) + \left(f(1) - f\left(1 - \frac{1}{n}\right)\right) n \left(t + \frac{1}{n} - 1\right)$, $1 - \frac{1}{n} \leq t \leq 1$.

Show that $d_p(g_n, f) \to 0$ for all $1 \leq p \leq \infty$.

Answer

Q2: A.34

Show that unions of open sets are open and intersection of any two open sets is open. Give an example to show that the intersection of an infinite number of open sets need not be open.

Answer

Q3: A.36

Let $f_n(x) = x^n$ and $g(x) \equiv 0$ on $\mathbb{R}$. Then $\{f_n\}_{n \geq 1}$ converges pointwise to g on $(-1,1)$, uniformly on $[a,b]$ for $-1 < a < b < 1$, but not uniformly on $(0,1)$.

Answer

Q4: 1.2

Let Ω be a finite set, i.e., the number of elements in Ω is finite. Let $\mathcal{F} \subset \mathcal{P}(\Omega)$ be an algebra. Show that $\mathcal{F}$ is a σ-algebra.

Answer

Q5: 1.4

Let $\{\mathcal{F}_\theta : \theta \in \Theta\}$ be a family of σ-algebras on Ω. Show that $\mathcal{G} \equiv \bigcap_{\theta \in \Theta} \mathcal{F}_\theta$ is also a σ-algebra.

Answer

Q6: 1.6

Let Ω be a nonempty set and let $\mathcal{A} \equiv \{A_i : i \in \mathbb{N}\}$ be a partition of Ω, i.e., $A_i \cap A_j = \emptyset$ for all $i \neq j$ and $\bigcup_{n \geq 1} A_n = \Omega$. Let $\mathcal{F} = \left\{\bigcup_{i \in J} A_i : J \subset \mathbb{N}\right\}$ where, for $J = \emptyset$, $\bigcup_{i \in J} A_i \equiv \emptyset$. Show that $\mathcal{F}$ is a σ-algebra.

Answer

Q7: Problem A

Let Ω be an uncountable set (e.g. $\mathbb{R}$) and $\mathbb{C}$ be the collection of all the singleton sets (i.e. $\mathbb{C} = \{\{\omega\} : \omega \in \Omega\}$). Find the smallest σ-algebra generated by $\mathbb{C}$. Justify your answer.

Answer